Independent variable
VS
Dependent variable

Independent variable

1. DEFINITION
2. KEY IDEA
3. OTHER NAMES

Independent variable

- Definition: The variable that you control, change or use as input to see its effect
- Key idea: It does not depend on another variable in the experiment
- Other names: 1- Feature (in ML)
  2- Predictor
  3- Input variable
  4- Explanatory variable
- Note: In Machine Learning(ML), independent variable are usually X (Features)

Dependent variable

1. DEFINITION
2. KEY IDEA
3. OTHER NAMES

Dependent variable

- Definition: The variable that changes depending on the independent variable
- Key idea: It is the result or outcome we measure
- Other names: 1- Target
  2- Label
  3- Output variable
  4- Response variable
- Note: In Machine Learning(ML), dependent variable are usually Y

Examples

CLEAR EXPLANATION

Simple example

- Suppose we study: Does studying more improving exam scores?
- Dataset

Student	Study Hours(X)	Exam Score(Y)
A	2	60
B	4	75
C	6	90

- 1- Independent variable(X): Study Hours
- 2- Independent variable(Y): Exam Score

- Because: Exam Score depend on Study Hours

In Machine Learning(ML)

- Dataset

Size	Rooms	Location	Price
120	6	Cairo	200000
200	4	Giza	350000

- 1- Independent variable(X): 1- Size
  2- Rooms
  3- Locations

- 2- Independent variable(Y): 1- Price

In Machine Learning(ML)

- Model learn the relationship:

1- Model Learns: Price = F(Size, Rooms, Location)

2- Price: Dependent variable (what you want to predict)

3- Size, Rooms, Location: Independent variable(predictors)

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- Simple explanations + Practical examples

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